LOST ENERGY

Game Design Document

Third-Person Adventure / Resource Management

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Engine : Unity 6 (URP)

Target Platform : Windows (PC)

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Main Cover Art Design

Figure 1. Lost Energy – Main Cover Art

Figure 2. *Alternative cover design supporting the game's atmospheric and narrative*

1. Design History

1.1 - Version 1.0

This version represents the submission-ready current design document for the Lost Energy project.

- Three playable scenes completed: SampleScene, SampleScene2, SampleScene3.
- Third-person character controller, orbit camera, and interaction system integrated.
- Oxygen-based resource management, crystal collection loop, and hazard zone system completed.
- Main menu, pause menu, settings panel, and loading screen implemented.
- NPC dialogue system, inter-scene transition flow, and basic audio management completed.
- Asset, source, and license information integrated into the document.

2. Game Overview

2.1 - Game Definition

Lost Energy is a single-player third-person adventure game. The player manages their oxygen to survive on a fragmented island floating in the void, collects crystals scattered throughout the environment, avoids hazardous zones, and opens gates that enable progression between scenes.

Figure 3. *General gameplay view showing the player character, HUD, and scene layout together.*

2.2 - Design Goals

- Establish gameplay that is quickly understood through simple rules yet creates pressure.
- Use the oxygen mechanic simultaneously as a life indicator, time pressure, and strategic resource.
- Combine environmental threats, exploration, and storytelling into a short but cohesive experience.
- Produce a finished-feeling vertical slice within a manageable scope for academic submission.

2.3 - Core Design Principles

2.3.1 - Constant Pressure

The player loses oxygen even in safe zones. As a result, inactivity is penalized, decision-making is accelerated, and every crystal becomes meaningful.

2.3.2 - Clear Objectives

In the first two scenes, the crystal target is explicit. The player can always see what they need to collect, when the door will open, and what the progression condition is.

2.3.3 - Environmental Storytelling

Red hazardous surfaces, scene transitions, island boundaries, and guardian dialogues work together to reinforce the broken structure of the world.

2.4 - Core FAQ

Question	Answer
What is the game?	A short-form third-person adventure built around resource management, exploration, and scene-based progression.
Where does it take place?	An island system that is part of an incomplete creation, floating in the void.
What does the player control?	A single character that moves, jumps, interacts, collects crystals, and progresses between scenes.
What is the main focus?	Choosing the right route under oxygen pressure, collecting crystals, and reaching the exit.
What makes it unique?	The oxygen system simultaneously determines time pressure, route optimization, and reward structure.

3. Feature Set

3.1 - Core Gameplay Features

- Third-person character controller
- Freely rotating orbit camera system
- Real-time oxygen consumption and replenishment system
- Crystal collection and objective completion structure
- IInteractable-based interaction architecture
- Additional oxygen loss from hazard zones
- Death/respawn flow tied to falling and oxygen depletion
- NPC dialogue system and scene-bound narrative delivery

3.2 - Supporting Systems

- Main menu and pause menu
- Settings panel and volume control
- Scene transition via loading screen
- Scene-based music switching
- VFX and SFX support for crystal collection
- Branching ending choice in the final scene

3.3 - User Experience Features

- Continuously visible oxygen and crystal indicators
- Prompt text on screen for nearby interaction targets
- Clear control scheme
- Compact level design suited to short play sessions

4. Game World

4.1 - General Structure

The Lost Energy world is designed as an incomplete realm of existence. The island is physically bounded; going beyond its borders results in death. Each new scene represents a zone where deterioration has become more visible.

4.2 - World Features

4.2.1 - Fragmented Island Structure

The play area is built on a layout where safe surfaces coexist with risky gaps. This makes navigation, staying on platforms, and selecting safe routes important.

4.2.2 - Deterioration Zones

Red hazardous surfaces do not merely provide visual variety; they are mechanical threat zones that force the player to alter their route.

4.3 - Physical World Summary

Parameter	Value
Perspective	Third-person (3D)
Walk Speed	10 units/sec
Sprint Speed	25 units/sec
Jump Height	1.5 units
Camera Distance	5 units

Table 1. *Core physical movement parameters governing the player character*

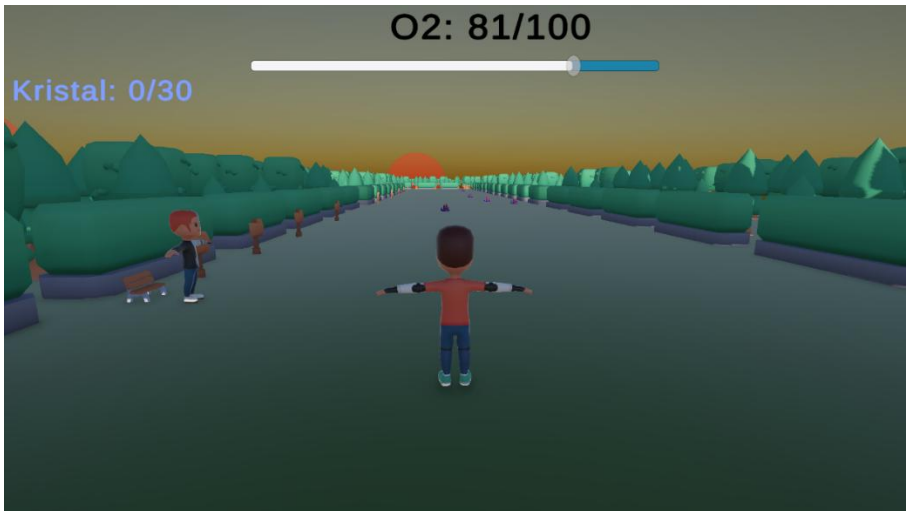

Figure 4. *General view of Scene 2 showing increased environmental deterioration.*

Figure 5. *General environment view belonging to the final scene (Scene 3).*

5. Camera and Controls

5.1 - Camera System

The camera system is managed within PlayerController3P. The camera orbits around the player; yaw and pitch inputs are handled independently. Since movement direction is calculated relative to the camera's view, the control feel is designed to resemble third-person action games.

Parameter	Value
Mouse Sensitivity	0.2
Pitch Minimum	-30°
Pitch Maximum	+60°
Camera Distance	5 units

Table 2. *Camera configuration values defined in `PlayerController3P`. Pitch limits prevent the camera from flipping at extreme angles: the −30° floor avoids disorientation on narrow platforms, while the +60° ceiling provides a generous upward view without exposing the empty void above the island. Mouse Sensitivity of 0.2 was tuned to feel responsive without over-rotating on typical mouse hardware.*

5.2 - Control Scheme

Action	Input
Move	WASD / Arrow Keys
Jump	Space
Sprint	Left Shift
Interact	E
Camera	Mouse
Pause / Release Cursor	ESC

Table 3. *Full player input mapping. All inputs are handled by Unity's new Input System package (`InputSystem_Actions`). The ESC key both pauses gameplay and releases/recaptures the cursor, removing the need for a separate cursor-toggle binding.*

6. Gameplay Systems

6.1 - Oxygen System

Oxygen is the game's primary resource structure. The player consumes oxygen as long as they are present in the scene. When the value reaches zero, the character dies and returns to the last safe starting point.

Zone	Starting Oxygen	Normal Consumption	Hazard Extra	Crystal Bonus
Scene 1	100	3 units/sec	+5 units/sec	+5
Scene 2	100	3 units/sec	+6 units/sec	+6
Scene 3	N/A	N/A	N/A	N/A

Table 4. *Per-scene oxygen parameters. Scene 3 has no crystal target and both exit doors are unlocked from the start; however, oxygen drain continues at the same base rate. The only pressure in Stage 3 is selection.*

6.2 - Crystal System

- Crystals are spawned in defined areas within the scene.
- The player collects a crystal by interacting with the E key.
- Visual and auditory feedback is generated during collection.
- In the first two scenes, completing the target crystal count opens the exit door.
- In the final scene there is no crystal target; the final choice is presented directly.

Each collected crystal replenishes +5 oxygen (Scene 1) or +6 oxygen (Scene 2), capped at the maximum of 100. At the 25-crystal target in Scene 1 the potential total replenishment is 125 oxygen - roughly two full oxygen cycles - which offsets approximately 42 seconds of normal drain. This balance is intentional: the reward is meaningful enough to incentivize collection but never fully eliminates pressure.

6.3 - Hazard Zone System

- HazardZone areas increase oxygen consumption.
- These zones are visually marked with red surfaces.
- When entering a hazard area, music and tension are intensified.

Figure 6. *Example screenshot showing the effect of a hazard zone on the play area.*

Figure 7. *Second view showing the pressure a hazard zone exerts on the player.*

6.4 - Interaction System

The interaction structure is built on the IInteractable interface. Crystals, doors, NPCs, and potions all use the same approach. PlayerInteraction identifies the most suitable target for the player and displays the appropriate prompt text on screen.

Figure 8. *The interaction prompt and HUD components visible during crystal collection.*

6.5 - Reward and Progression Loop

```
Collect Crystal --> Replenish Oxygen --> Survive Longer
                --> Collect More Crystals --> Complete Objective
                --> Open Exit Door --> Transition to Next Scene
```

7. World Layout and Progression Flow

7.1 - General Flow

The game consists of three main scenes. In the first two scenes the player collects a set number of crystals; in the third scene the experience ends or loops through a final choice.

7.2 - Scene 1 – Island Entrance

Field	Description
Atmosphere	Relatively calm starting zone
Crystal Target	25
Hazards	Falling off island edges, low-density hazard zones
NPC	Guardian
Exit	ExitDoorSceneLoader → Scene 2

Table 5. *Scene 1 layout and objective summary.*

7.3 - Scene 2 – Deteriorated Zone

Field	Description
Atmosphere	Increased deterioration with more aggressive risk areas
Crystal Target	30
Hazards	Falling off island edges, dense hazard zones, stronger route pressure
NPC	Guardian
Exit	ExitDoorSceneLoader → Scene 3

Table 6. *Scene 2 layout and objective summary.*

7.3 - Scene 3 – End of Creation

Field	Description
Atmosphere	The final threshold of an unfinished existence
Crystal Target	0
Hazards	Final decision: loop or exit choice
NPC	Guardian
Exit	FinalDoorController → Loop end or Win screen

Table 7. Scene 3 layout and objective summary. This scene has no crystal requirement; both doors are open from the start.

7.4 - Game Flow Diagram

```
Main Menu
 --> LoadingScreen
 --> Scene 1  (Crystal Target: 25)
 --> Scene 2  (Crystal Target: 30)
 --> Scene 3  (No crystal target)
     --> Loop Door  : Return to Scene 1
     --> Exit Door  : Congratulations Screen
```

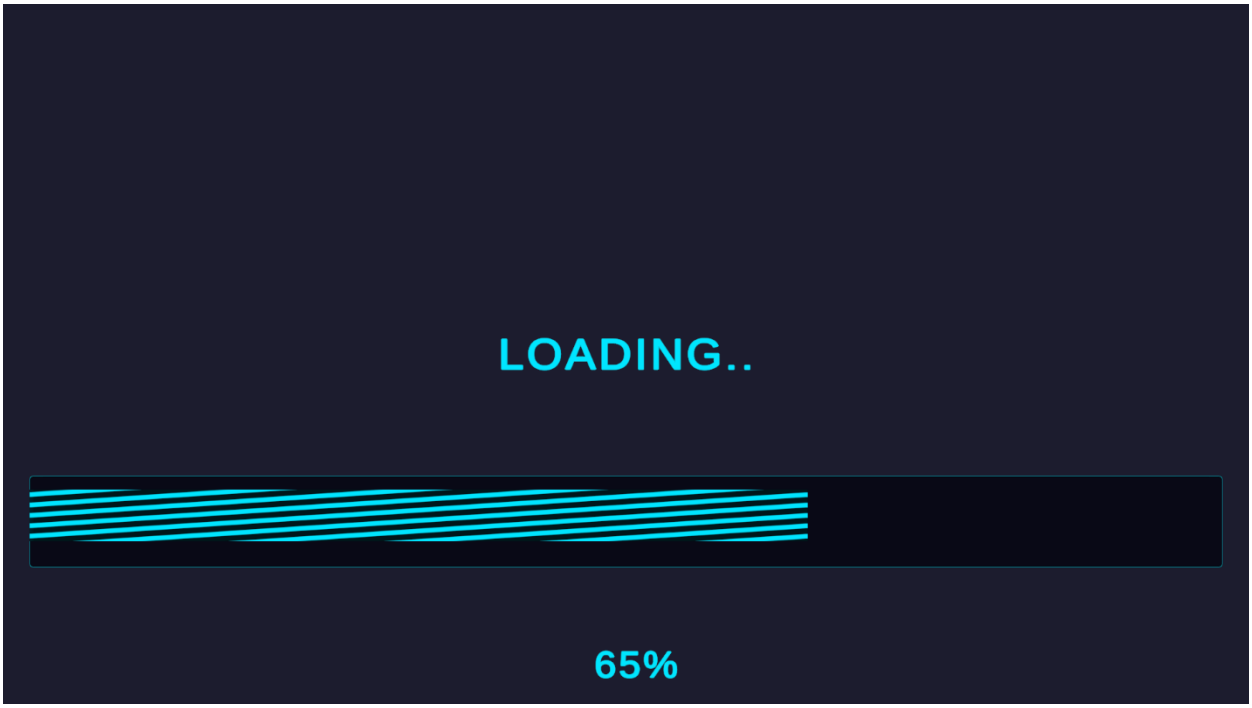

Figure 9. *The loading screen is used for transitions between scenes.*

Figure 10. *The loop-return door interaction in the final scene.*

8. Characters

8.1 - Player Character

Attribute	Description
Model	Kenney Animated Characters Protagonists
Control Setup	PlayerController3P + CharacterController
Primary Goal	Collect crystals and reach the exit under oxygen management
Death Conditions	Oxygen reaching zero or falling so that Y < -1
Narrative Background	A figure trapped within an unfinished creation whose reason for existing is unclear. Initially reactive and survival-driven - a stranger to the system. By Scene 3, capable of a conscious choice about their place within the cycle.

Table 8. *Player character properties and narrative role.*

8.2 - NPC – Guardian

The Guardian is the game's sole narrative-focused NPC. It greets the player in each scene, provides information about the current zone, and gradually reveals the world's background. Because the character has merged with the island, it cannot leave - a fact that reinforces its narrative function.

8.3 - Dialogue System Behavior

- Managed through NPCDialogue and DialogueData structures.
- Oxygen consumption is paused while dialogue is active.
- Dialogue simultaneously provides system tutorials and story delivery.

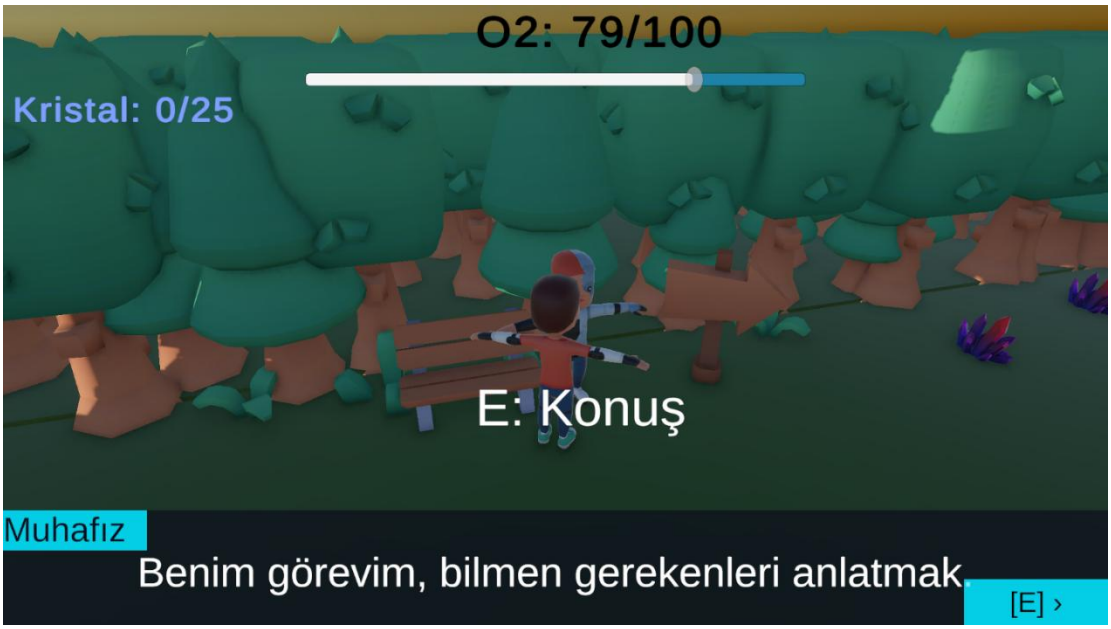

Figure 11. *The dialogue panel that opens during interaction with the Guardian character.*

9. User Interface

9.1 - HUD Components

Component	Function
Oxygen Slider	Visually displays the current oxygen level
Oxygen Value	Shows the numerical equivalent of oxygen
Crystal Counter	Shows collected crystal count in current / target format
Interaction Prompt	Provides contextual information for nearby interactable objects

***Table 9**. HUD components permanently visible during gameplay. All elements update in real time and are managed by `UIManager`.*

9.2 - Panel Structure

Panel	Trigger Condition
Game Over	Oxygen depletion or death by falling
Win	Selecting the winning exit in the final scene
Pause	ESC key
Settings	Accessed from within the Pause menu
Controls	Accessed from the main menu or Pause menu
Loading Screen	Opens automatically during scene transitions

***Table 10.** In-game panel inventory and their activation conditions. All panel states are managed through `UIManager` and `PauseManager`.*

9.3 - Menu Structure

The main menu contains options to start the game, access settings, view controls, and quit. At game start Time.timeScale = 1f and the cursor is free.

***Figure 12.** The game's main menu interface.*

Figure 13. *The UI screen showing the game's control scheme.*

Figure 14. *The settings panel containing volume level and mute options.*

10. Audio and Music

10.1 - Audio Architecture

The project uses three primary channels through GameAudioMixer.

Channel	Purpose
MasterVol	Overall level of all sounds
MusicVol	Background music tracks
SFXVol	In-game effects, dialogue audio, and hazard warnings

Table 11. Audio mixer channel structure. All three channels are exposed as parameters in `GameAudioMixer` and controlled at runtime by `SettingsManager` using logarithmic dB conversion. Volume preferences are persisted via `PlayerPrefs`.

10.2 - Music Usage

- Different music streams are used for the main menu and gameplay scenes.
- Hazard zones are supported with additional audio layers that heighten tension.
- MusicManager maintains music continuity across scene transitions.

10.3 - Source Summary

Source	Content
Flowerhead – SomeWhatGood: Lofi	Observatory and chill 2
Not Jam Music Pack	ChillMenu, CriticalTheme, SwitchWithMeTheme
Freesound	Crystal collection and NPC sound effects

Table 12. *Music track sources used across the project's scenes and menus.*

11. Single-Player Experience

11.1 - Game Loop

The single-player experience is built on a scene-based structure that creates short but intense decision pressure. In each scene the player reads the environment, evaluates risky routes, collects crystals, and reaches the exit door.

11.2 - Estimated Playtime

Play Style	Duration
Speed Completion	5–6 minutes
Exploration and Dialogue Focus	10–12 minutes

Table 13. *Estimated completion time by playstyle*

11.3 - Win and Lose Conditions

State	Condition
Win	Selecting the exit ending via FinalDoorController in Scene 3
Lose	Oxygen reaching zero or falling from the scene

Table 14*. Win and lose conditions. There is no partial failure state; any death immediately triggers the respawn flow managed by `RespawnManager`.*

Figure 15. *The exit door interaction offering the option to complete the game.*

Figure 16. *The result screen shown after the game is completed.*

12. Story and Narrative

12.1 - Story Summary

Lost Energy's narrative is delivered almost entirely through three conversations with the Guard - the sole human presence in the game world. Across three encounters the Guard progressively reveals the true nature of the island: it is not a real world but an unfinished creation, abandoned mid-construction by a god who was still learning. The oxygen instability, the red deterioration zones, the void at the island's edges - all are symptoms of an incomplete existence.

The player character arrives without memory or explanation, navigates the fragmenting island, and ultimately reaches a binary choice: step through the exit door and attempt to escape the cycle, or return through the loop door and forget everything once more. The Guard cannot make this journey - he is part of the world, condemned to remember every iteration of the loop while the player perpetually forgets.

Crystals carry a thematic weight beyond their mechanical function: they are described by the Guard as the only "solidly built" fragments within the broken creation - pieces of genuine reality preserved within the decay. Collecting them is not merely a survival act; it is the act of anchoring oneself to what little is real.

12.2 - Thematic Structure

Element	Meaning
Island	An incomplete creation
Oxygen	The world's unstable and incomplete nature
Crystals	The only solidly built fragments - preserved pieces of genuine reality
HazardZone	Physical manifestation of spreading deterioration
Guard	A witness bound to the system, unable to leave it
Loop Door	The choice to return, forget, and begin again
Exit Door	The choice to confront an uncertain outside

Table 15. Narrative symbolism mapping each game element to its thematic meaning within the world of Lost Energy.

12.3 - Scene-Based Narrative Development

Scene	Narrative Function
Scene 1	World rules introduced; player as unknowing arrival; god as distant but present force
Scene 2	Decay revealed as spreading; god reframed as inexperienced; existential unease escalated
Scene 3	Full truth delivered: incomplete creation, loop existence, Guard's limitation, player's choice

***Table 16**. Per-scene narrative delivery functions. Each scene's dialogue builds directly on the previous, forming a three-act revelation structure.*

12.3 - Character Motivations

Player character: Arrives without explanation and operates on survival instinct for the first two scenes. In Scene 3 they are offered the full truth and must decide whether to act on it or surrender to the loop. They are silent - they cannot speak - which the Guard explicitly acknowledges mid-dialogue in Scene 1.

The Guard: Locked into his role as a part of the world he describes. He guides out of function, not hope. He remembers every cycle the player has forgotten, and carries the weight of that knowledge alone.

12.4 - Full Guard Dialogue

The Guard's dialogue is triggered once per scene via NPCDialogue when the player interacts with him. Oxygen consumption is paused for the entire duration. The scripts below document all lines from DialogueData.json, DialogueData2.json, and DialogueData3.json, organized into narrative beats. Beat labels are documentation groupings - they are not tagged in the asset files.

12.4.1 - Scene 1 – Island Entrance (DialogueData.json)

Narrative function: Introduce the player to the world's rules, the oxygen system, crystals as exit tickets, and the island's boundary. Close with the first hint of the world's deeper philosophical nature.

Beat 1 - The Guard stops the player

#	Turkish	English
1	Dur!	Stop!
2	Gitmeden önce bilmen gereken birkaç şey var.	There are a few things you need to know before you go.

Beat 2 - Existential ambivalence about the player's presence

#	Turkish	English
3	Buraya nasıl geldin bilmiyorum.	I don't know how you got here.
4	Neden geldin onu da bilmiyorum.	I don't know why you came either.
5	Açıkçası umurumda da değil.	Frankly, I don't care.
6	Ama gelmişsen...	But if you came here...
7	Tanrının bir bildiği vardır.	God must know something.

Beat 3 - Mission: deliver the briefing

#	Turkish	English
8	Benim görevim, bilmen gerekenleri anlatmak.	My duty is to tell you what you need to know.

Beat 4 - The island and the void

#	Turkish	English
9	İlk olarak...	First of all...
10	Hiçliğin ortasında süzülen bir adadayız.	We are on an island floating in the middle of nothingness.
11	Ormanların içine girmeni pek tavsiye etmem.	I don't recommend going into the forests.
12	Ada sınırına fazla yaklaşırsan...	If you get too close to the island's edge...
13	Hiçliğe düşersin.	You'll fall into the void.

Beat 5 - Oxygen bar tutorial

#	Turkish	English
14	Şimdi asıl konuya gelelim.	Now let's get to the main point.
15	Kafanın üstünde gördüğün şey...	What you see above your head...
16	O bir oksijen barı.	That's an oxygen bar.
17	Bitmesini isteyeceğini sanmıyorum.	I don't think you'd want it to run out.
18	Ama denemek istersen...	But if you want to try...
19	Sonuçlarını ben de merak ediyorum.	I'm curious about the consequences myself.

Beat 6 - Navigation

#	Turkish	English
20	Çıkışa gidene kadar okları takip et.	Follow the arrows until you reach the exit.

Beat 7 - Crystal tutorial

#	Turkish	English
21	Kristalleri toplamayı da unutma.	Don't forget to collect the crystals.
22	Onlar senin çıkış biletin.	They are your exit ticket.
23	Biraz da can verirler.	They'll give you a little life too.

Beat 8 - Creator's nature (first hint)

#	Turkish	English
24	Tanrımızın garip huyları vardır.	Our god has strange habits.
25	Ama bazen insaflıdır.	But sometimes he's merciful.

Beat 9 - The player cannot speak

#	Turkish	English
26	Ee...	Well...
27	Sormak istediğin bir şey var mı?	Is there something you want to ask?
28	...	...
29	Ah...	Ah...
30	Siz konuşamıyorsunuz.	You can't speak.
31	Unutuyorum bazen.	I forget sometimes.

Beat 10 - Shared nature (existential close)

#	Turkish	English
32	Bizim tek farkımız...	Our only difference...
33	Biraz daha fazla farkında olmamız.	is that we're a little more aware.
34	Ama sonuçta...	But in the end...
35	Aynı yerin parçalarıyız.	we are all parts of the same place.

Closing

#	Turkish	English
36	İleride daha çok şey öğreneceksin.	You will learn much more ahead.
37	Dediklerimi unutma.	Don't forget what I've told you.
38	Bol şans, yabancı.	Good luck, stranger.

12.4.2 - Scene 2 – Deteriorated Zone (DialogueData2.json)

Narrative function: Acknowledge the player's progress; explain hazard zones as corrupted areas; reframe the god as a new, inexperienced creator causing involuntary damage; escalate existential unease.

Beat 1 - Recognition

#	Turkish	English
1	Hey...	Hey...
2	Yine sen.	It's you again.
3	Buraya kadar gelebildin demek.	So you managed to get this far.
4	İlk aşamayı geçtin.	You passed the first stage.

Beat 2 - Omitted warning

#	Turkish	English
5	Sana geçen sefer söylemeyi unutmuştum.	I forgot to tell you something last time.
6	Ama fark etmişsindir.	But you must have noticed.

Beat 3 - Hazard zones explained

#	Turkish	English
7	Bazı bölgelerde oksijen daha hızlı bitiyor.	Oxygen runs out faster in some areas.
8	Neden mi?	Why?
9	Çünkü bu yer... düzgün çalışmıyor.	Because this place... doesn't work properly.
10	Fark ettiysen, zemin de kırmızı.	If you noticed, the floor is also red.
11	Bu iyiye işaret değil.	That's not a good sign.
12	Oralar daha çok bozulmuş alanlar.	Those are more corrupted areas.

Beat 4 - God's experiments spreading the decay

#	Turkish	English
13	Tanrı yine bir şeyler denerken...	While God was trying things again...
14	Bir şeyleri bozmuş.	He broke something.
15	Fark etmişsindir...	You must have noticed...
16	Şimdi de alanlar büyümüş.	Now the areas have grown too.
17	Bu da bozulmanın yayıldığı anlamına geliyor.	This means the corruption is spreading.

Beat 5 - Warning

#	Turkish	English
18	Dikkatli ol.	Be careful.

Beat 6 - Remaining rules unchanged

#	Turkish	English
19	Geri kalan şeyler aynı.	The rest remains the same.
20	Hızlı ol.	Be quick.
21	Kristalleri topla.	Collect the crystals.
22	Kapıya ulaş.	Reach the door.

Beat 7 - Unstable world, ominous luck wish

#	Turkish	English
23	Bol şans.	Good luck.
24	Bu bölgede buna ihtiyacın olacak.	You'll need it in this area.
25	Bu ada...	This island...
26	Hiç stabil sayılmaz.	Can't be called stable at all.

Beat 8 - God as a newborn creator

#	Turkish	English
27	Neyse...	Anyway...
28	Fazla konuşmayayım.	I shouldn't talk too much.
29	Sağı solu belli olmuyor.	Nothing is predictable anymore.
30	Galiba bu evrende yeni.	He seems to be new to this universe.
31	Her şeyi öğrenmeye çalışıyor.	Trying to learn everything.
32	Tıpkı yeni doğmuş bir bebek gibi.	Just like a newborn baby.
33	Bu yüzden bu anomaliler.	That's why these anomalies.

Beat 9 - Uncertain farewell

#	Turkish	English
34	Neyse...	Anyway...
35	Bu bölgeyi geçersen...	If you pass through this area...
36	Yine karşılaşırız.	We'll meet again.
37	Tabii...	Of course...
38	Orası da varsa.	If that place still exists.

Closing

#	Turkish	English
39	Görüşmek üzere.	Until we meet again.
40	Güç seninle olsun.	May the force be with you.
41	İhtiyacın olacak.	You'll need it.

12.4.3 - Scene 3 – End of Creation (DialogueData3.json)

Narrative function: Deliver the complete truth - the world is an unfinished creation; crystals are fragments of reality; the loop is real; the Guard cannot leave; the player has a choice.

Beat 1 - Arrival acknowledged

#	Turkish	English
1	Ah...	Ah...
2	Sonunda geldin.	You finally made it.
3	Buraya kadar gelen pek olmaz.	Not many get this far.

Beat 2 - Truth begins

#	Turkish	English
4	Şimdi...	Now...
5	Gerçeği öğrenme zamanı.	It's time to learn the truth.

Beat 3 - The world is unfinished

#	Turkish	English
6	Burası bir dünya değil.	This is not a world.
7	Yarım kalmış bir yaratım.	An unfinished creation.
8	Tanrının denemesi.	God's experiment.
9	Ama tamamlanamamış.	But it was never completed.
10	O yüzden her şey bozuk, çirkin ve orantısız.	That's why everything is broken, ugly, and disproportionate.
11	O yüzden oksijen azalıyor.	That's why oxygen diminishes.
12	O yüzden hiçlik var.	That's why the void exists.
13	Bu yer silinmedi.	This place was not deleted.
14	Ama düzeltilmedi de.	But it wasn't fixed either.
15	Sadece bırakıldı.	It was simply left behind.

Beat 4 - Crystals redefined as fragments of reality

#	Turkish	English
16	Kristaller...	Crystals...
17	Onlar bu yerin sağlam inşa edilmiş yegane parçaları.	They are the only solidly built parts of this place.
18	Gerçekliğin kırıntıları.	Fragments of reality.
19	Topladıkça...	As you collect them...
20	Buradan koparsın.	You detach from this place.
21	Ve belki çıkarsın.	And perhaps you escape.

Beat 5 - Doubt about the exit

#	Turkish	English
22	Ama çıkış...	But the exit...
23	Gerçek mi bilmiyorum.	I don't know if it's real.
24	Belki başka bir deneme.	Perhaps another experiment.

Beat 6 - Staying means decay

#	Turkish	English
25	Ama burada kalmak...	But staying here...
26	Onu biliyorum.	That I know.
27	Burası çürüyor.	This place is rotting.

Beat 7 - The loop revealed

#	Turkish	English
28	Ve evet...	And yes...
29	Bu bir döngü.	This is a loop.
30	Sen daha önce de buradaydın.	You have been here before.
31	Ama hatırlamıyorsun.	But you don't remember.
32	Ben hatırlıyorum.	I remember.

Beat 8 - The Guard's nature revealed

#	Turkish	English
33	Ben çıkamıyorum.	I cannot leave.
34	Ben bu yerin parçasıyım.	I am part of this place.
35	Ama sen...	But you...
36	Henüz değilsin.	Not yet.

Beat 9 - The binary choice

#	Turkish	English
37	Seçimin var.	You have a choice.
38	Devam et.	Continue.
39	Ya da kal.	Or stay.
40	Ve unut.	And forget.
41	...	...

Beat 10 - The doors pointed out

#	Turkish	English
42	Kapılar orada.	The doors are there.

Closing

#	Turkish	English
43	Bu sefer son olabilir.	This time could be the end.
44	Ya da sadece başka bir başlangıç.	Or just another beginning.
45	Bol şans.	Good luck.
46	Bu defa gerçekten ihtiyacın olacak.	This time you'll really need it.

Table 17. *Full Guard dialogue scripts for all three scenes. Original Turkish text with English translation. Beat groupings are documentation labels - they are not tagged in the DialogueData ScriptableObject assets. Delivered via NPCDialogue + DialogueManager; oxygen drain is paused for the full duration of each conversation.*

13. Technical Implementation

13.1 - Architecture Summary

```
LostEnergy Namespace
├── GameManager
├── MusicManager
├── SettingsManager
├── SceneLoader
└── UIManager
```

13.2 - Core Scripts

Script	Responsibility
PlayerController3P	Character movement, sprint, jumping, and orbit camera
OxygenSystem	Oxygen consumption and replenishment
GameManager	Crystal counting, objective checking, and death flow
CrystalCollectible	Crystal interaction, SFX and VFX triggering
HazardZone	Applying additional oxygen consumption
DialogueManager	Dialogue flow and oxygen pause
RespawnManager	Position reset after death
PauseManager	Pause flow and panel management
LoadingScreenManager	Asynchronous scene loading and loading screen

Table 18. Core script inventory and their primary responsibilities. Scripts communicate through C# events (`UnityAction`, `Action<T>`) rather than direct references, keeping coupling low.

13.3 - Technical Decisions

- **Modular script structure:** Each system (oxygen, crystals, hazards, dialogue) is encapsulated in its own script and communicates via C# events. This makes individual systems independently testable and allows DialogueManager to pause OxygenSystem without any coupling to GameManager.
- **MusicManager uses DontDestroyOnLoad:** This prevents music from restarting or cutting out during asynchronous scene loading. The alternative - restarting music from Awake() in each scene - was evaluated but produced an audible silence gap during transitions that broke immersion.
- **SettingsManager uses PlayerPrefs:** Volume preferences persist between play sessions without requiring a save file. A ScriptableObject-based approach was considered but rejected as unnecessarily complex for three simple float values.
- **References cached in Start():** Components are located once at initialization and stored in private fields. Calling GetComponent<T>() every frame inside Update() at 60 fps would add unnecessary overhead across all active scripts.

13.4 - Performance Notes

- The crystal VFX structure uses an Instantiate + Destroy approach, sufficient for low object density.
- Scene scales are limited for short gameplay sessions, keeping overall runtime cost manageable.

14. Assets and License Information

14.1 - License Summary

Asset Group	License / Usage Terms
Kenney Assets	CC0 1.0 Universal
Unity Asset Store Assets	Standard Unity Asset Store EULA
Not Jam Music Pack	CC0 1.0
Flowerhead – SomeWhatGood: Lofi	Royalty free (author declaration)
Freesound effects	Per-file CC license
Unity built-in packages	Unity Asset Store EULA / Unity Companion License

Table 19. High-level license summary by asset source group.

14.2 - Kenney Assets

All Kenney assets are published under the CC0 1.0 Universal license (https://creativecommons.org/publicdomain/zero/1.0/).

Asset Pack	Version	Source
Animated Characters Protagonists	1.1	kenney.nl/assets/animated-characters
Fantasy Town Kit	2.0	kenney.nl/assets/fantasy-town-kit
Graveyard Kit	5.0	kenney.nl/assets/graveyard-kit
Nature Kit	2.1	kenney.nl/assets/nature-kit
Platformer Kit	4.1	kenney.nl/assets/platformer-kit

Table 20. Kenney asset packs used in the project. All are published under CC0 1.0 Universal - no attribution required.

14.3 - Unity Asset Store Assets

Asset Pack	Publisher	Version
Stylized Crystal	LowlyPoly	1.0
Stylized door	lowpoly89	1.0

Table 21. Unity Asset Store assets used in the project.

License: Unity Asset Store End User License Agreement – unity.com/legal/as-terms

14.4 - Audio and Music Sources

File / Pack	Source	License
794489__gobbe57__coin-pickup.wav	freesound.org/people/gobbe57/	Freesound per-file
822698__metris__retro-npc-voice.wav	freesound.org/people/Metris/	Freesound per-file
ChillMenu.wav	not-jam.itch.io/not-jam-music-pack	CC0 1.0
CriticalTheme.wav	not-jam.itch.io/not-jam-music-pack	CC0 1.0
SwitchWithMeTheme.wav	not-jam.itch.io/not-jam-music-pack	CC0 1.0
Observatory and chill 2.wav	flowerheadmusic.itch.io/somewhat-good-lofi	Royalty free

Table 22. *Individual audio and music file sources. Files obtained from Freesound require per-file license verification; CC BY files require written attribution.*

14.5 - Unity Built-in Packages

Package	Publisher	License
TextMesh Pro	Unity Technologies	Unity Asset Store EULA
Input System	Unity Technologies	Unity Companion License
Post Processing Stack	Unity Technologies	Unity Companion License

Table 23. Unity built-in packages included via the Package Manager.

14.6 - Usage Notes

- Usage terms for Unity Asset Store assets are covered under the standard Unity EULA.
- For files sourced from Freesound, the license type must be verified per file.
- If a CC BY licensed Freesound file is used, written attribution is required.

15. Installation and Running

15.1 - Development Environment Requirements

- Unity 6 (recommended: 6000.0.x LTS)
- Universal Render Pipeline (URP)
- Input System package

15.2 - Minimum System Requirements

Component	Specification
OS	Windows 10 64-bit
CPU	Intel Core i3 class or equivalent
Memory	4 GB RAM
GPU	DirectX 11 compatible graphics card
Storage	At least 1 GB free space
Resolution	1280 x 720

Table 24. Minimum system requirements for running the compiled Windows build.

15.2 - Build Steps

1. Open File → Build Settings inside the Unity Editor.
2. Verify that scenes are listed in the following order:

```
0 – MainMenu
1 – LoadingScreen
2 – SampleScene
3 – SampleScene2
4 – SampleScene3
```

3. Select Windows, x86_64 as the platform.
4. Use Build and select Builds/Windows/ as the output folder.

15.3 - Running the Game

The compiled build is launched via the Builds/Windows/Lost Energy.exe file.

15.4 - Quick Control Reference

```
WASD        -> Move
Left Shift  -> Sprint
Space       -> Jump
E           -> Interact / Collect Crystal
Mouse       -> Camera
ESC         -> Pause Menu
```